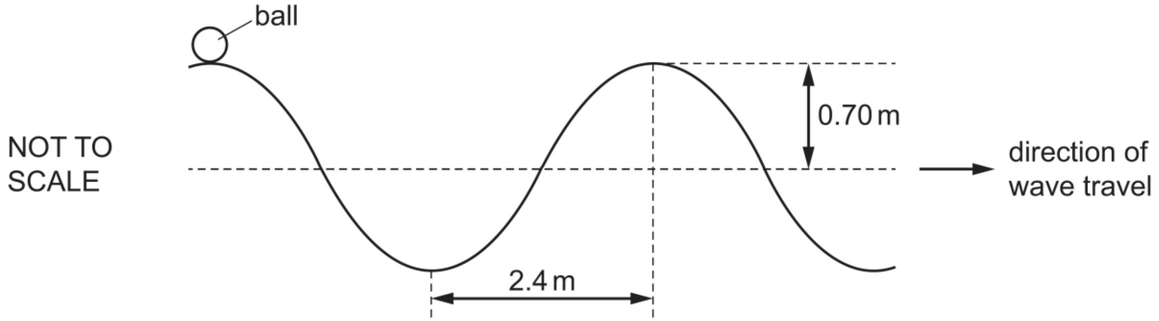
16	A transverse water wave is moving along the surface of some water. This causes a ball to move vertically without moving horizontally as it floats upon the surface. At one instant, the ball is at the position shown.



The wave has a frequency of 0.20 Hz and an amplitude of 0.70 m. The distance between a trough and an adjacent peak is 2.4 m.

What is the distance travelled by the ball in a time of 20 s?

	A	5.6 m	**B**	9.6 m	**C**	11.2 m	**D**	19.2 m

Answer: C

Period of the wave = $\frac{1}{f}$ = 5 s

In 20 s, the ball would have oscillated by $\frac{20}{5}$ = 4 cycles

It would have moved a total distance of 4 x (0.7 x 4) = 11.2 m

Students who found the speed of the wave using fλ, and then take speed x time = dist would end up getting 19.2 m (D).

17 A sound wave travels from left to right across a room. The variation with distance across the